MAC-1G_AMBA

Gigabit Ethernet Media Access Controller

Tests set details

Version 1.00

Contents

Tables

1. General informations

All tests are grouped into few categories with respect to their functionality. Although it is possible to change operating frequencies of MAC-1G_AMBA core via generic assignments, all test files are prepared for use with the default settings.

Each test is composed of fourteen files:

/apbstim.txt	: Stimulus file for APB interface
/smemstim.txt	: Stimulus file for shared memory model
/arbstim.txt	: Stimulus file for AMBA™ arbiter model
/linkstim.txt	: Stimulus file for GMII/MII interface
/apbcomp.txt	: Expected Test Bench results from APB interface.
/ahbcomp.txt	: Expected Test Bench results from AHB interface.
/linkcomp.txt	: Expected Test Bench results from GMII/MII interface.
/intcomp.txt	: Expected Test Bench results from interrupt line.
/apbdiff.txt	: APB interface difference file
/ahbdiff.txt	: AHB interface difference file
/linkdiff.txt	: GMII/MII interface difference file
/intdiff.txt	: Interrupt line difference file.
/time.txt	: Proposed simulation time.
/generic.txt	: Set of generic parameters

For detailed information about testbench configuration refer to MAC-1G_AMBA Media Access Controller Specification.

2. Backoff/Deferring tests

Tests in this group are prepared to test compatibility with IEEE 802.3 standard for Carrier Sense Multiple Access with Collission Detection (CSMA/CD).

Table 1. Backoff/Deferring tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	1 collision	bd1	
2	2 collisions	bd2	
3	3 collisions	bd3	
4	15 collisions	bd4	
5	16 collisions	bd5	
6	late collision	bd6	
7	normal and late collisions	bd7	
8	collision during preamble	bd8	
9	collision during sfd	bd9	
10	collision during crc	bd10	
11	deferring	bd11	

3. Transmit FIFO tests

Tests in this group are prepared to test transmit FIFO operation.

Table 2. Transmit tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	long frame transmission	tfifo1	frame 1 : size = 8192 frame 2 : size = 8191 frame 3 : size = 8194
2	long frame transmission	tfifo2	frame 1 : size = 2047 frame 2 : size = 2047 frame 3 : size = 2047 frame 4 : size = 2047
3	long frame transmission	tfifo3	frame 1 : size = 8192 frame 2 : size = 8192 frame 3 : size = 8192 frame 4 : size = 8192 frame 5 : size = 8192
4	treshold levels: TR=1, TTM=0	tfifo4	frame 1 : size = 2047 frame 2 : size = 2047 frame 3 : size = 2047 frame 4 : size = 2047
5	treshold levels: TR=2, TTM=0	tfifo5	as above
6	treshold levels: TR=3, TTM=0	tfifo6	as above
7	treshold levels: TR=0, TTM=1	tfifo7	as above
8	treshold levels: TR=1, TTM=1	tfifo8	as above
9	treshold levels: TR=2, TTM=1	tfifo9	as above
10	treshold levels: TR=3, TTM=1	tfifo10	as above
11	treshold levels: TR=0, TTM=0, SF=1	tfifo11	as above
12	underflow condition	tfifo12	frame 1 : size = 2047und. frame 2 : size = 2047 frame 3 : size = 2047und. frame 4 : size = 2047und. frame 5 : size = 2047

4. Transmit linked list state machine tests

Tests in this group are prepared to test linked list state machine which is part of MACs Descriptor/Buffer architecture.

Table 3. Transmit linked list state machine tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	1 buffer, various frame sizes for address(2..0)=000, for buf1	tlsm1	
2	1 buffer, various address alignments for size(2..0)=000, for buf1	tlsm2	
3	1 buffer, various frame sizes for address(2..0)=000, for buf2	tlsm3	
4	1 buffer, various address alignments for size(2..0)=000, for buf2	tlsm4	
5	1 buffer, various frame sizes for address(2..0)=001, for buf1	tlsm5	
6	1 buffer, various frame sizes for address(2..0)=010, for buf1	tlsm6	
7	1 buffer, various frame sizes for address(2..0)=011, for buf1	tlsm7	
8	1 buffer, various frame sizes for address(2..0)=100, for buf1	tlsm8	
9	1 buffer, various frame sizes for address(2..0)=101, for buf1	tlsm9	
10	1 buffer, various frame sizes for address(2..0)=110, for buf1	tlsm10	
11	1 buffer, various frame sizes for address(2..0)=111, for buf1	tlsm11	
12	1 buffer, various frame sizes for address(2..0)=001, for buf2	tlsm12	
13	1 buffer, various frame sizes for address(2..0)=010, for buf2	tlsm13	
14	1 1 buffer, various frame sizes for address(2..0)=011, for buf2	tlsm14	
15	1 buffer, various frame sizes for address(2..0)=100, for buf2	tlsm15	
16	1 buffer, various frame sizes for address(2..0)=101, for buf2	tlsm16	
17	1 buffer, various frame sizes for address(2..0)=110, for buf2	tlsm17	
18	1 buffer, various frame sizes for address(2..0)=111, for buf2	tlsm18	
19	2 buffers, various buffer 1 sizes for buffer 2 address(2..)="000"	tlsm19	

20	2 buffers, various buffer 1 sizes for buffer 2 address(2..)="001"	tlsm20	
21	2 buffers, various buffer 1 sizes for buffer 2 address(2..)="010"	tlsm21	
22	2 buffers, various buffer 1 sizes for buffer 2 address(2..)="011"	tlsm22	
23	2 buffers, various buffer 1 sizes for buffer 2 address(2..)="100"	tlsm23	
24	2 buffers, various buffer 1 sizes for buffer 2 address(2..)="101"	tlsm24	
25	2 buffers, various buffer 1 sizes for buffer 2 address(2..)="110"	tlsm25	
26	2 buffers, various buffer 1 sizes for buffer 2 address(2..)="111"	tlsm26	
27	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)="000"	tlsm27	
28	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)="001"	tlsm28	
29	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)="010"	tlsm29	
30	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)="011"	tlsm30	
31	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)="100"	tlsm31	
32	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)="101"	tlsm32	
33	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)="110"	tlsm33	
34	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)="111"	tlsm34	
35	2 descriptors/3 buffers per frame	tlsm35	
36	2 descriptors/4 buffers per frame	tlsm36	
37	3 descriptors per frame	tlsm37	
38	7 descriptors per frame	tlsm38	
39	7 descriptors per frame, chain mode	tlsm39	
40	empty descriptors	tlsm40	

5. Transmit interrupt mitigation control tests

Tests in this group are prepared to test transmit interrupt mitigation control.

Table 4. Transmit interrupt mitigation control tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	TT=1, NRP=0, CC=1	tim1	
2	TT=1, NRP=7, CC=1	tim2	
3	TT=0, NRP=7, CC=1	tim3	
4	TT=1, NRP=7, CC=1, IC=1	tim4	
5	TT=2, NRP=0, CC=1	tim5	
6	TT=3, NRP=0, CC=1	tim6	
7	TT=14, NRP=0, CC=1	tim7	
8	TT=15, NRP=0, CC=1	tim8	
9	TT=2, NRP=0, CC=0	tim9	
10	TT=3, NRP=0, CC=0	tim10	
11	TT=0, NRP=1, CC=1	tim11	
12	TT=0, NRP=2, CC=1	tim12	
13	TT=0, NRP=3, CC=1	tim13	
14	TT=0, NRP=4, CC=1	tim14	

6. Receive operation tests

Tests in this group are prepared to test receive operation.

Table 5. Receive operation tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	ethernet frame with 1517 bytes	rc1	
2	ethernet frame with 1519 bytes	rc2	
3	802.3 frame with 1517 bytes	rc3	
4	802.3 frame with 1519 bytes	rc4	
5	frame with 8193 bytes	rc5	
6	pass bad frame mode - frame with 63 bytes	rc6	
7	frame with crc error	rc7	
8	frame with data alignment error	rc 8	
9	frame with mii error	rc 9	
10	receiving in the suspended/stopped state	rc 10	

7. Receive address filtering tests

Tests in this group are prepared to test receive address filtering.

Table 6. Receive address filtering tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	perfect filtering	raf1	
2	perfect filtering – filtering fail	raf2	
3	inverse perfect filtering	raf3	
4	hash filtering	raf4	
5	hash only filtering	raf5	
6	receive all mode	raf6	
7	promiscous mode	raf7	
8	pass all multicast mode	raf8	
9	external address filtering	raf9	

8. Receive FIFO tests

Tests in this group are prepared to test receive FIFO operation.

Table 7. Receive FIFO operation tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	various frame sizes	rfifo1	frame 1 : size = 68 frame 2 : size = 69 frame 3 : size = 70 frame 4 : size = 71 frame 5 : size = 72 frame 6 : size = 73 frame 7 : size = 74 frame 8 : size = 75
2	various frame sizes	rfifo2	frame 1 : size = 1516 frame 2 : size = 1517 frame 3 : size = 1518 frame 4 : size = 1519 frame 5 : size = 1520 frame 6 : size = 1521 frame 7 : size = 1522 frame 8 : size = 1523
3	various frame sizes	rfifo3	frame 1 : size = 2044 frame 2 : size = 2045 frame 3 : size = 2046 frame 4 : size = 2047 frame 5 : size = 2048 frame 6 : size = 2049 frame 7 : size = 2050 frame 8 : size = 2051
4	various frame sizes	rfifo4	frame 1 : size = 8190 frame 2 : size = 8191 frame 3 : size = 8192 frame 4 : size = 8193 frame 5 : size = 8194 frame 6 : size = 8195 frame 7 : size = 8196 frame 8 : size = 8197
5	various frame sizes	rfifo5	frame 1 : size = 2048 frame 2 : size = 2048 frame 3 : size = 2048 frame 4 : size = 2048 frame 5 : size = 2048 frame 6 : size = 2048 frame 7 : size = 2048 frame 8 : size = 2048

6	overflow condition	rfifo6	frame 1 : size = 8190ove. frame 2 : size = 8190 frame 3 : size = 8190ove. frame 4 : size = 8190ove. frame 5 : size = 8190

9. Receive linked list state machine tests

Tests in this group are prepared to test receive linked list state machine which is part of MACs Descriptor/Bufer architecture.

Table 8. Receive operation tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	single buffer, various frame sizes for address(2..0)=111, for buf2	rlsm1	
2	2 buffers per frame	rlsm2	
3	2 descriptors/2 buffers per frame	rlsm3	
4	2 descriptors/3 buffers per frame	rlsm4	
5	2 descriptors/4 buffers per frame	rlsm5	
6	3 descriptors per frame	rlsm6	
7	7 descriptors per frame	rlsm7	
8	7 descriptors per frame, chain mode	rlsm8	
9	empty descriptors	rlsm 9	
10	descriptors unavailable	rlsm10	

10. Receive interrupt mitigation control tests

Tests in this group are prepared to test receive interrupt mitigation control.

Table 9. Receive interrupt mitigation control tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	RT=3, NRP=0, CC=1	rim1	
2	RT=4, NRP=7, CC=1	rim2	
3	RT=0, NRP=7, CC=1	rim3	
4	RT=4, NRP=0, CC=1	rim4	
5	RT=5, NRP=0, CC=1	rim5	
6	RT=14, NRP=0, CC=1	rim6	
7	RT=15, NRP=0, CC=1	rim7	
8	RT=2, NRP=0, CC=0	rim8	
9	RT=3, NRP=0, CC=0	rim9	
10	RT=0, NRP=1, CC=0	rim10	
11	RT=0, NRP=2, CC=0	rim11	
12	RT=0, NRP=3, CC=0	rim12	

11. Interrupt tests

Tests in this group are prepared to test interrupts.

Table 10. Interrupt tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	early receive interrupt	int1	
2	receive interrupt	int2	
3	receive buffer unavailable interrupt	int3	
4	receive process stopped interrupt	int4	
5	early transmit interrupt	int5	
6	transmit interrupt	int6	
7	transmit buffer unavailable interrupt	int7	
8	transmit process stopped interrupt	int8	

12. General purpose timer tests

Tests in this group are prepared to test general purpose timer.

Table 11. General purpose timer tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	TIM=65535, CON=0	gpt1	
2	TIM=1024, CON=0	gpt 2	
3	TIM=1023, CON=0	gpt 3	
4	TIM=1022, CON=0	gpt 4	
5	TIM=3, CON=0	gpt 5	
6	TIM=2, CON=0	gpt 6	
7	TIM=1, CON=0	gpt 7	
8	TIM=65535, CON=1	gpt 8	
9	TIM=1024, CON=1	gpt 9	
10	TIM=1023, CON=1	gpt 10	
11	TIM=1022, CON=1	gpt 11	
12	TIM=3, CON=1	gpt 12	
13	TIM=2, CON=1	gpt 13	
14	TIM=1, CON=1	gpt 14	

13. DMA tests

Tests in this group are prepared to test Direct Memory Access operation.

Table 12. DMA tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	burst length = 0, arbitration mode 1	dma1	
2	burst length = 1, arbitration mode 1	dma2	
3	burst length = 2, arbitration mode 1	dma3	
4	burst length = 4, arbitration mode 1	dma4	
5	burst length = 8, arbitration mode 1	dma5	
6	burst length = 16, arbitration mode 1	dma6	
7	burst length = 32, arbitration mode 1	dma7	
8	burst length = 0, arbitration mode 2	dma8	
9	burst length = 1, arbitration mode 2	dma9	
10	burst length = 2, arbitration mode 2	dma10	
11	burst length = 4, arbitration mode 2	dma11	
12	burst length = 8, arbitration mode 2	dma12	
13	burst length = 16, arbitration mode 2	dma13	
14	burst length = 32, arbitration mode 2	dma14	
15	big endian mode for descriptors	dma15	
16	big endian mode for buffers	dma16	
17	big endian mode for descriptors/buffers	dma17	
18	wait cycles for burst length = 0	dma18	
19	wait cycles for burst length = 1	dma19	

14. Miscellaneous tests

Tests in this group are prepared to test functionality not covered by functional tests.

Table 13. Miscellaneous tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	transmission with clkdma,clkcsr = clkmii/10	misc1	
2	transmission with clkdma,clkcsr = 10*clkmii	misc2	
3	transmission with clkdma = clkmii = 10*clkcsr	misc3	
4	transmission with clkdma = clkmii = clkcsr/10	misc4	
5	receiving with clkdma,clkcsr = clkmii/10	misc5	
6	receiving with clkdma,clkcsr = 10*clkmii	misc6	
7	receiving with clkdma = clkmii = 10*clkcsr	misc7	
8	receiving with clkdma = clkmii = clkcsr/10	misc8	
9	simultanous transmission/receiving with clkdma,clkcsr = clkmii/10	misc9	
10	simultanous transmission/receiving with clkdma,clkcsr = 10*clkmii	misc10	
11	simultanous transmission/receiving with clkdma = clkmii = 10*clkcsr	misc11	
12	simultanous transmission/receiving with clkdma = clkmii = clkcsr/10	misc12	